

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA17

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### COURSE OUTCOME COVERED

**CO1:** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

*Assessment Tool Weightage: 50%*

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**QUESTIONS: 5**

**Duration : 1 Hour**  
**Total Marks:50**

Annexure A

### Analytical Problem Solving

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#### ***Code of Conduct:***

*I G. Gowthami/192524285 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

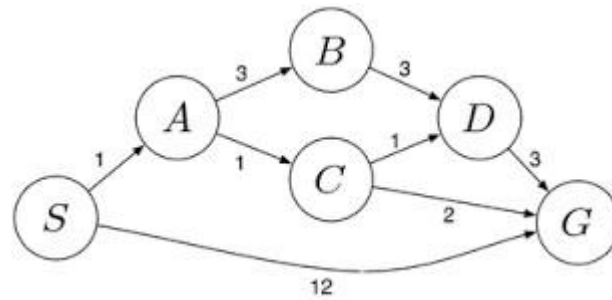
***Signature of the student:G.Gowthami***

## Annexure A

### **Analytical Problem Solving**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
  - i. What are the percept's for this agent?
  - ii. Characterize the operating environment.
  - iii. What are the actions the agent can take?
  - iv. How can one evaluate the performance of the agent?
  - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



## ANSWERS

### 1. Water Jug Problem (4-Gallon and 3-Gallon Jugs)

Steps to get exactly 2 gallons in the 4-gallon jug:

Fill the 3-gallon jug completely. (3,0)

Pour it into the 4-gallon jug. (0,3)

Fill the 3-gallon jug again. (3,3)

Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full. Only 1 gallon is needed, so 2 gallons remain in the 3-gallon jug. (2,4)

Empty the 4-gallon jug. (2,0)

Pour the 2 gallons from the 3-gallon jug into the 4-gallon jug. (0,2)

Result: The 4-gallon jug contains exactly 2 gallons.

### 2. Mars Rover Agent

i. What are the percepts for this agent?

Camera images

Rock and soil samples

Temperature readings

Atmospheric pressure

Terrain information

Obstacle detection

GPS/position data

ii. Characterize the operating environment

Partially Observable

Deterministic

**Sequential**

**Dynamic**

**Continuous**

**Single Agent**

**iii. What are the actions the agent can take?**

**Move forward/backward**

**Turn left/right**

**Capture images**

**Collect soil/rock samples**

**Analyze samples**

**Send data to Earth**

**Avoid obstacles**

**iv. How can one evaluate the performance of the agent?**

**Accuracy of collected data**

**Number of successful samples collected**

**Distance covered**

**Energy efficiency**

**Mission completion rate**

**Safe navigation**

**v. Suitable agent architecture**

**Utility-Based Agent**

**Reason: It selects actions that maximize scientific value while minimizing energy use and avoiding risks.**

**3. 8-Queens Problem**

**Problem Components**

**Initial State**

**Empty 8×8 chessboard.**

**State Space**

**All possible arrangements of queens on the board.**

**Operators**

**Place one queen in a valid position of the next column.**

**Goal State**

**Place all 8 queens so that no two queens attack each other.**

**Goal Test**

**No two queens are in the same:**

**Row**

**Column**

**Diagonal**

**Path Cost**

**Each move has equal cost (1).**

**Search Strategies**

**Breadth First Search (BFS)**

**Depth First Search (DFS)**

**Backtracking (most commonly used)**

**Hill Climbing**

**A\* Search**

#### **4. OLA Cab Booking Application**

**Type of Problem-Solving Agent**

**Goal-Based Agent**

**Reason: The agent chooses the most suitable cab type based on the user's goal (destination, comfort, fare, etc.).**

**Pseudocode**

**START**

**Input pickup location**

**Input destination**

**Display available cab types**

**User selects cab**

**IF cab available THEN**

**Calculate fare**

**Confirm booking**

**Assign driver**

```
    Display driver details
ELSE
    Display "Cab not available"
ENDIF

END
```

## **5. Uniform Cost Search (UCS) – Logistics Company**

### **Uniform Cost Search Algorithm**

**Start from warehouse S.**

**Add S to the priority queue with cost 0.**

**Remove the node with the lowest cost.**

**If it is the goal G, stop.**

**Otherwise, expand its neighboring warehouses.**

**Add neighbors with updated cumulative costs.**

**Continue until G is reached.**

**Return the least-cost path.**

**Pseudocode**

**UniformCostSearch(Start, Goal)**

**Create Priority Queue**

**Insert Start with cost 0**

**WHILE** queue is not empty

**Remove** node with minimum cost

**IF** node == Goal

**Return** path

**Expand** neighbors

**FOR** each neighbor

**Calculate total cost**  
**Insert into queue**  
**END FOR**

**END WHILE**

**Return Failure**

**Advantages**

**Finds the optimal (least-cost) path.**

**Suitable for weighted graphs.**

**Complete and optimal when all costs are positive.**

**RUBRICS FOR CALCULATION:**

| Criteria                  | Weightage | Level 4<br>(Exemplary<br>)                                                        | Level 3<br>(Proficient<br>)               | Level 2<br>(Developing<br>)                | Level 1<br>(Beginning<br>)                    |
|---------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection          | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |
| Solution Process          | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              |
| Accuracy of Calculation   | 20%       | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |
| Interpretation of Results | 15%       | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |



|                       |     |                                                           |                                     |                                            |                                               |
|-----------------------|-----|-----------------------------------------------------------|-------------------------------------|--------------------------------------------|-----------------------------------------------|
| Problem Understanding | 20% | Clearly interprets problem, identifies all parameters and | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
|                       |     | requirements accurately                                   |                                     |                                            |                                               |